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Exploring the Influence of Coherent Breathing on Psychophysiological Stress During a Simulated 3-Day 400-m Race

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Abstract

Chiron, F, Gulören, B, Erblang, M, Poirier, C, Elbaz, M, Servonnet, A, Gaudin, M, Le Coz, C, Lopes, P, Chennaoui, M, Hanon, C, Léger, D, and Thomas, C. Exploring the influence of coherent breathing on psychophysiological stress during a simulated 3-day 400-m race. *J Strength Cond Res* 40(5): e460–e471, 2026—Managing psychophysiological stress is essential for elite athletes, especially during the high-intensity competitive period. Coherent breathing has been proposed as a noninvasive strategy to enhance recovery by modulating autonomic, neuroendocrine, and inflammatory responses. This study investigated the effects of coherent breathing on the autonomic nervous system (ANS), hypothalamic-pituitary-adrenal (HPA) axis regulation, inflammatory response, and sleep parameters in well-trained athletes undergoing repeated supramaximal efforts. Twenty-two athletes ($N = 22$) were randomly assigned to a control group (CONT, $n = 11$) or a coherent breathing group (RELAX, $n = 11$). Over an 8-day protocol simulating competition, athletes completed 3 400-m races. Heart rate variability, salivary biomarkers (alpha-amylase, cortisol, testosterone IL1- β), and sleep parameters were assessed throughout this study. Two-way repeated measure ANOVA revealed that the RELAX group showed significantly increased parasympathetic activity (root mean square of successive difference, $p < 0.01$) and a more stable cortisol response ($p < 0.01$) compared with the control group. The RELAX group also exhibited lower IL1- β levels ($p < 0.05$), longer total sleep duration ($p < 0.05$), and reduced sleep latency ($p < 0.05$). No significant differences in athletic performance were observed between groups. Coherent breathing effectively modulates psychophysiological stress by enhancing ANS balance, regulating HPA axis activity, reducing inflammatory responses, and improving sleep quality. This accessible and low-cost intervention may support recovery and resilience in athletes exposed to repeated high-intensity efforts.

Key Words: heart rate variability, salivary samples, well-trained athletes, sleep regulation, controlled breathing

Introduction

One of the primary challenges faced by elite athletes is managing the anxiety-provoking external conditions often associated with competitive environments and repeated maximal performances. Stress, in this context, refers to physiological and psychological demands that exceed the organism's typical adaptive capacity, resulting in a disruption of homeostasis (44), a response commonly observed during supramaximal exercise (29). Psychophysiological stress, characterized by a combination of cognitive and physiological responses, can be defined as a state of actual or perceived threat that alters the body's equilibrium (46). Such stress responses can manifest through behavioral (43), psychological (3), and physiological (47) changes. In addition, stress-induced alterations commonly affect biological systems, including the dysregulation of inflammatory processes, which

may increase susceptibility to various pathophysiological conditions (42).

Numerous studies have already documented the impact of supramaximal exercise and competitive stress on homeostatic balance, particularly with respect to the autonomic nervous system (ANS) and the hypothalamic-pituitary-adrenal (HPA) axis (17,23,24,40). More specifically, psychophysiological stress caused by consecutive supramaximal exercises in elite athletes has been shown to attenuate parasympathetic modulation and disrupt salivary hormone profiles, which are key indicators of systemic homeostasis. The combined effects of intense physical exertion and competitive psychological pressure significantly perturb autonomic and neuroendocrine functions, leading to imbalances that can compromise both performance and recovery (20). Accordingly, the regulation of psychophysiological stress remains a critical concern for elite athletes performing at international levels.

Recovery, broadly defined as the set of physiological and psychological processes that restore homeostasis and replenish the athlete's capacity for performance following training or competition, is considered particularly important in events where athletes must compete repeatedly within a limited time frame (34). However, many elite athletes do not always prioritize their recovery strategies. Indeed, the sometimes unpredictable nature of

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competition may prevent athletes from adhering to recommendations aimed at mitigating stress and supporting recovery, especially in terms of sleep management, despite their awareness of the impact of insufficient or poor-quality sleep on these processes (48). Sleep is recognized by both the high-performance community (coaches and athletes) and the scientific community as the most essential recovery modality for optimizing physical and mental performance, yet it is also highly susceptible to stress-induced disruptions (14). Furthermore, the causal links between psychophysiological stress and sleep disturbances can be highlighted through quantitative sleep analysis, such as sleep quality or sleep latency, as well as qualitative analysis by referring to different sleep stages, particularly the slow-wave sleep phase (14).

In this context, the implementation of structured breathing strategies may serve as a valuable tool for regulating psychophysiological stress (45), optimizing sleep, and ultimately enhancing postexercise recovery. Among these strategies, slow-paced breathing, often referred to as coherent breathing, has gained attention due to its cost-effectiveness, accessibility, and ease of application in athletic populations (4). Beyond its simplicity, coherent breathing has demonstrated a range of beneficial physiological and psychological effects aimed at supporting both performance and recovery (36). Specifically, coherent breathing stimulates neurophysiological processes that modulate autonomic balance, endocrine function, and inflammatory responses. These include the reactivation of parasympathetic activity following high-intensity exercise, which is associated with improved sleep parameters and accelerated physical restoration (13). By activating the afferent vagus nerve, coherent breathing contributes to neuroendocrine regulation (25), immune modulation (2), and attenuation of anxiety symptoms (27), all of which are critical components of effective recovery. In addition, coherent breathing may improve sleep quality not only by reducing latency and promoting longer duration, but also by facilitating deeper stages of sleep, such as slow-wave sleep, which are essential for muscle repair and hormonal restoration. Therefore, breathing interventions such as coherent breathing represent a promising recovery modality for managing stress and promoting physiological resilience during demanding competition periods (35). However, to date, no study has examined the specific effects of a controlled breathing protocol on integrated biomarkers of the ANS and the HPA axis in the context of repeated supramaximal efforts in well-trained athletes.

The objective of this study was to evaluate the effects of a coherent breathing protocol on psychophysiological stress using a multifactorial design, including biomarkers of ANS activity and HPA axis function in well-trained athletes exposed to a simulated competition. Before integrating this breathing method into elite-level athletic routines, our aim was to determine whether coherent breathing could attenuate stress responses and modulate biological indicators of homeostasis, while also improving sleep during repeated supramaximal efforts. We specifically hypothesized that coherent breathing would enhance parasympathetic regulation, reduce inflammatory and neuroendocrine markers reflected in salivary biomarkers, and improve recovery-related sleep metrics, including reduced sleep latency and increased total sleep duration, thereby supporting performance across a sequence of 3 400-m races.

Methods

Experimental Approach to the Problem

We implemented a simulated competition designed to closely replicate the conditions and effects of a real competitive event.

Given the psychophysiological stress responses typically observed in world-class and elite athletes during actual competitions, this interventional protocol aimed to regulate psychophysiological stress during a series of 400-m races in a simulated competition. The data collected, based on the same psychophysiological markers as in the previous study and a biomarker of inflammatory stress, combined with sleep monitoring, allowed us to verify whether coherent breathing can help regulate stress and thereby improve recovery in athletes specialized in supramaximal exercises. To compare the measurements obtained during the simulated competition, the baseline was established over the 3 days preceding the testing phase, while the posttest period corresponded to the 2 days following the last race (Figure 1).

Subjects

In accordance with previous studies (17), to achieve a statistical power of 95% with a 5% significance level, 22 ($n = 22$: 15 men: 22.8 ± 3.3 years, 67.2 ± 7.5 kg, 178 ± 6.9 cm and 7 women: 21.3 ± 2.4 years, 58.2 ± 4.4 kg, 170 ± 5.6 cm) highly trained athletes (national level) from the French Athletics Federation were included in this study. According to the classification by McKay et al., these highly trained athletes had personal best (PB) performances that met the criteria for national-level competition (50.44 seconds for men and 59.14 seconds for women in the 400 m; 56.24 seconds for men and 67.04 seconds for women in the 400-m hurdles; and 177.00 seconds for men and 140.00 seconds for women in the 800 m), as indicated in the Hungarian table of the World Athletics Quotation Table (WA) for the 400 m, 400-m hurdles, or 800-m events. The Hungarian table is internationally recognized as the most accurate performance scoring table in world athletics and is certified by World Athletics for ranking athletes based on their performance level. All subjects were informed about the study protocol, their rights, and the associated risks before providing written informed consent. All procedures were approved by the CERSTAPS ethics committee (CERSTAPS No. 2022-A00644-39/approval date March 15, 2022) and were conducted in accordance with the Declaration of Helsinki (1964, revised in 2001).

Procedures

The subjects ($N = 22$) completed their first visit (day 0: familiarization), during which the initial measurements were taken. The preliminary assessments included the Competitive State Anxiety Inventory-2 (CSAI-2R) questionnaire, a micro blood sample (for lactate levels), and instructions for the collection of salivary markers, sleep data, and ANS data (heart rate variability [HRV]). Written informed consent was also obtained from the subjects. The subjects were then randomly assigned to 1 of 2 groups: either the control group (CONT, $n = 11$) or the group practicing a relaxation technique known as coherent breathing (RELAX, $n = 11$). Subsequently, subjects independently continued (pretest: days 1, 2, and 3) to collect all psychophysiological stress markers, with the RELAX group implementing their relaxation strategy 3 times per day. During the test period, subjects ran a 400m race on three consecutive days, with 24 hours of recovery between each race, simulating a series of heats, each consisting of two athletes with similar personal bests, in order to challenge the athletes. The semifinals and finals of a major championship (test: days 4, 5, and 6). All psychophysiological stress markers were collected during this period. After the 3 test days, subjects continued independently while continuing to collect psychophysiological stress markers (posttest: day 7).

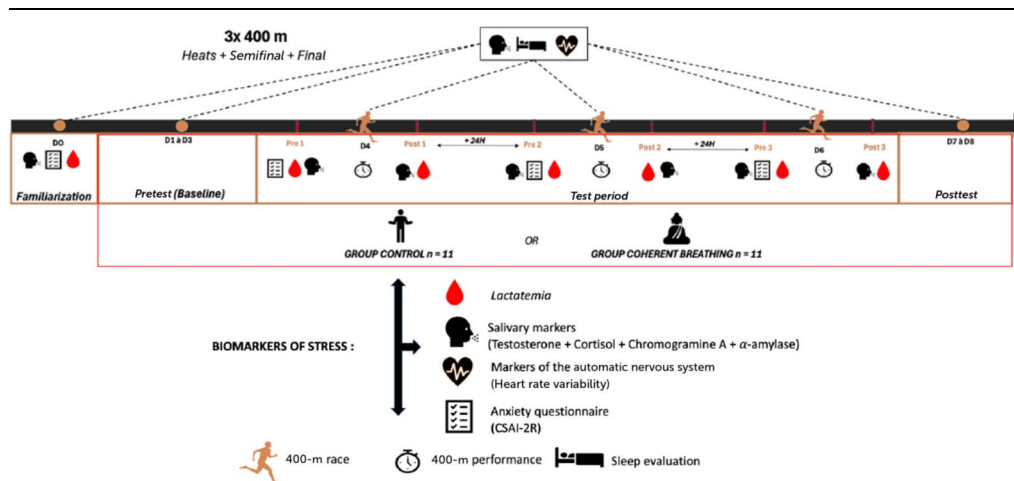


Figure 1. Experimental design of the protocol during the simulated competition. ANS markers, salivary biomarkers, and sleep data were collected throughout the protocol. Blood lactate levels and salivary biomarkers were measured before and after each 400-m race during the testing period. The CSAI-2R was realized during the familiarization day as well as before each 400-m race during the testing period.

Strategy of Recovery Application (Coherent Breathing With Respi Relax⁺). Among the 22 subjects, 11 were randomly assigned to form the RELAX condition ($n = 11$). They were instructed to practice coherent breathing 3 times a day (in the morning after waking and saliva sampling, after lunch, and just before bedtime). Coherent breathing is a gentle relaxation method based on breathing, which helps reduce stress by regulating heart rate (HR). It adheres to the 365 rule: 3 times a day; 6 breaths per minute, which corresponds to a 5-second inspiration phase followed by a 5-second expiration phase, for 5 minutes. The exercise was to be performed while seated, with a straight back, feet flat on the ground, and legs uncrossed to free the abdomen. Inhalation was to be deep and regular, reaching the abdomen, with each inhalation and exhalation lasting 5 seconds, resulting in 6 breathing cycles per minute. Subjects were to perform the exercise using the RespiRelax application (available on Android and Apple stores, Allevard, France), which assists in regulating breathing (inhalation and exhalation) during the 5-minute exercise with the help of a ball that follows the rhythm corresponding to the correct breathing frequency. Subjects had a familiarization session with the experimenters during the initial visit to ensure that they could practice independently.

The 400-m Races. During the testing phase (day 3, day 4, and day 5), the subjects competed in a daily 400-m race. To ensure having the same baseline levels for every athlete, different internal controls were regulated before the experiment, such as training volume and intensity, with athletes being outside of intense specific preparation periods and stopping training during the experiment. In addition, the food intake was regulated through a daily resume of the quantity (Potential Renal Acid Load) and content of the meals, which did not differ between groups. These races served as simulated competitions, including a specific warm-up, a call room, and a direct confrontation race with a start from starting blocks. For performance measurements (split times on the 400 m in seconds), video recordings were made using an iPhone 11 (1080p HD at 60 frames per second, Apple, Cupertino, CA) and an iPad Pro (1080p HD at 60 frames per second, Apple). A hand clap was used to start the timer. The video camera was positioned perpendicularly to the finish line, similar to official competitions, and the timer was

stopped when the athlete's shoulder line crossed the finish line. The footage was analyzed using software to obtain precise measurements to the 100th of a second. In addition, a manual stopwatch was used as a backup measure in case of failure of the primary method. All performance tests were recorded by dual manual timing by 2 official athletics judges, specialists from the French national 400-m team, who took all race times simultaneously to avoid any methodological bias (17).

The 400-m races were conducted on an indoor track with a length of 340 m (Maigrot Hall of the Institute of Sport Expertise and Performance, Paris, France) to ensure highly reproducible conditions (the temperature was kept constant, and there was no wind). Before each race, the athletes prepared as if for a real competition, using techniques to stay calm and focused. The athletes were conditioned by their coaches as in a real competition context. Training loads were adjusted and reduced in the days leading up to the tests to ensure they were in optimal shape and motivated to perform. The athletes' warm-up lasted 1 hour, as in competition. The athletes were required to adhere to their usual competition routine, which included mobilization, jogging, stretching, technical exercises, sprints, and starts, before entering the call room. Tactically, the athletes were specifically trained to execute a quick and efficient start using their phosphocreatine reserves (38) to gain an initial advantage. As 400-m specialists, they had also learned to manage their effort with a fast start and maintain their resistance to the inevitable decline in speed that occurs in the last 100 m (28). Furthermore, the subjects engaged in direct 1-on-1 confrontations with instructions to win the race and achieve the best possible performance, as if they had to qualify for the next round, with their ranking influencing their participation the following day. Throughout the tests, the athletes received verbal encouragement and support, like in a competitive environment, to ensure they felt supported and motivated.

Lactatemia. Blood lactate concentration ([La]) was measured by fingertip puncture (20 μ L) to assess acid-base regulation. Samples were collected during the initial visit (familiarization) at rest (before starting the warm-up) and at the end of each race, 4 minutes (+3') and 8 minutes (+6') postexercise. The samples were analyzed using a Lactate Pro 2 device (Arkray, LT-1730, Kyoto, Japan).

Competitive State Anxiety Inventory-2 Questionnaire. A modified version of the CSAI-2R was used to measure cognitive anxiety, somatic anxiety, and self-confidence before performance (37). The CSAI-2 includes 27 items, with 9 items in each subscale. Examples of cognitive anxiety items include “I am concerned about this competition” and “I am worried about performing poorly,” while somatic anxiety items include “I feel nervous” and “My body feels tense.” The response scale allowed subjects to rate the intensity of each symptom on a scale from 1 (not at all) to 4 (very much), resulting in scores ranging from 9 to 36 for each subscale. In addition, a direction scale was included for the cognitive and somatic anxiety items, where subjects rated the extent to which the intensity of each symptom facilitated or debilitated subsequent performance on a scale from -3 (very debilitating) to +3 (very facilitating). Thus, the possible direction scores for each subscale ranged from -27 to +27, with a positive score indicating a facilitating state and a negative score indicating a debilitating state. Each subject completed the CSAI-2R questionnaire during the pretest period (familiarization), establishing a baseline value. Subsequently, all 44 subjects completed the questionnaire before each race and before starting their warm-up. The athletes were isolated to avoid disturbances during the questionnaire completion, ensuring reproducible conditions.

Heart Rate Variability. Lying to Standing Test: Orthostatic Test. The orthostatic test, a variant of the tilt test, aimed to collect R-R intervals over 10 minutes, alternating 5 minutes in a lying position with 5 minutes in a standing position. This test was to be conducted at the same time each day (~30 minutes) in a dark room (i.e., curtains drawn) (6). Before starting the test, athletes were instructed to use the bathroom to avoid sympathetic activation during recording. Upon waking, subjects attached a Polar H10 heart rate strap (H10, Polar Electro GmbH, Finland) around their chest and remained lying on their bed for 5 minutes before standing up and remaining still for 5 minutes. Subjects were required to breathe normally and spontaneously. Data were recorded continuously throughout the 10-minute test through Bluetooth using the Elite HRV app (version 5.2.1, Elite HRV Inc., Asheville, NC), which allows recording of the most common HRV indices. All data were then downloaded into Kubios HRV Premium software (version 3.4.3) for signal analysis.

Analysis of Vagally Mediated Heart Rate Variability. After data export, each data file was manually inspected to correct artifacts and then analyzed using specialized analysis software “Kubios HRV Standard” (The Biomedical Signal and Medical Imaging Analysis Group, Department of Applied Physics, University of Kuopio, Finland). A systematic mean correction was applied to all data to reduce the number of artifacts. A test was considered unusable if the percentage of artifacts was greater than 5%. Indices such as the average HR (HR: an indicator of overall ANS activation) and the root mean square of successive differences (RMSSD: an indicator of vagal activity) were calculated and considered as indices of the parasympathetic branch of the ANS.

Salivary Samples. *Saliva Collection.* First, salivary samples were collected during the morning of each period, right after the HRV testing. Moreover, prerace and postrace samples have been collected during the simulated competition period. The prerace sample has been collected before warming up, and the postrace sample has been collected 10 minutes after the end of the exercise. Samples were collected 3 days before the test period (during the pretest period to establish baseline levels) and during the 2 days

following the competition (posttest period) to assess the recovery kinetics. Saliva had to be collected at least 30 minutes after the last drink, food, and oral rinse. All athletes were in good health and had no history of somatic or sleep disorders. To standardize saliva collection, subjects were seated in a comfortable position. A minimum of 3 ml has been collected for the different salivary samples and spares. After the collection, samples have been aliquoted in a 3-ml microtube, which has been transferred directly to a -20° C container, and were stocked later in a -80° C freezer. Some of the saliva samples were subject to a simple freeze-thaw cycle to aliquot these samples. During the concentration measurements, samples were centrifuged (14,000×g, 2 minutes) to remove particulate matter, and clear samples were transferred into appropriate test wells. All analyses were conducted in our analysis laboratory as part of routine processing.

Alpha-Amylase. Commercial enzymatic assay kits (IBL-Tecan, Hamburg, Germany) were used to measure salivary alpha-amylase. All samples were measured in singlicate, and all data were expressed as absolute concentrations. As specified by the manufacturer, samples were diluted at a ratio of 1/301. Salivary alpha-amylase data are expressed in arbitrary units per milliliter (U·ml⁻¹). During singlet determinations, the application domain of salivary cortisol measurement was between 0 and 400 U·ml⁻¹. Interassay and intraassay coefficients of variation for the measurements are in the range specified by the manufacturer.

Cortisol. Commercial ELISA kits (Salimetrics LLC, State College, PA) were used to measure salivary cortisol. All samples were measured in singlicate, and all data were expressed as absolute concentrations. Samples were dosed pure during measurement. Salivary cortisol data are expressed in micrograms per deciliter (µg·dl⁻¹). During singlet determinations, the quantification limit of salivary cortisol was 0.012 µg·dl⁻¹. Interassay and intraassay coefficients of variation for the measurements are in the range specified by the manufacturer.

Testosterone. Commercial ELISA kits (Salimetrics LLC) were used to measure salivary testosterone. All samples were measured in singlicate, and all data were expressed as absolute concentrations. Samples were dosed pure during measurement. Salivary testosterone values are expressed in picograms per milliliter (pg·ml⁻¹) but converted to micrograms per deciliter (µg·dl⁻¹) for T/C ratio calculation. During singlet determinations, the quantification limit of salivary testosterone was 6.1 pg·ml⁻¹. Interassay and intraassay coefficients of variation for the measurements are in the range specified by the manufacturer.

IL1-β. Commercial ELISA kits (Salimetrics LLC) were used to measure salivary IL1-β. All samples were measured in singlicate, and all data were expressed as absolute concentrations. As specified by the manufacturer, samples were diluted at a ratio of 1/15. Salivary IL1-β data are expressed in picograms per milliliter (pg·ml⁻¹). During singlet determinations, the quantification limit of salivary IL1-β was 3.13 pg·ml⁻¹. Interassay and intraassay coefficients of variation for the measurements are in the range specified by the manufacturer.

Qualitative and Quantitative Sleep Monitoring: Oura Ring Version 3. The Oura Ring Version 3 (Oura Health Ltd., Finland) is a commercial device designed for tracking activity, sleep, HR, and HRV. The Oura Ring includes an integrated photoplethysmography sensor and an inertial measurement unit. The ring is lightweight (4–6 g), waterproof, and user-friendly, with a battery life of 6 days. Data are automatically transmitted to the mobile application (compatible with Android and iOS) and then

transferred to the cloud server. Subjects were required to wear the Oura Ring each night from the initial visit (familiarization) until the end of the test (D7). Data were automatically updated every morning and exported from the cloud, where data were collected every 5 seconds. The sleep metrics analyzed included total sleep duration, sleep latency, time spent in bed, duration of slow-wave sleep, duration of light sleep, and duration of rapid eye movement (REM) sleep.

Statistical Analyses

All values are presented as means $\pm$ SDs. The Shapiro-Wilk test was used to assess the normality of the data distribution. An a priori power analysis indicated that 22 subjects per condition were needed to detect significant differences, based on an estimated alpha level of 0.05 and a power of 95%, following performance improvement data from a previous study by Chiron et al. (17).

A 2-way repeated measures ANOVA was used to interpret the results. Post hoc tests were also applied to compare specific measures between the 2 competition days. Greenhouse-Geisser correction was applied to the 2-way repeated measures ANOVA results.

The analysis covered different testing periods (baseline, heats, finals, postcompetition). The alpha level was set at $p \leq 0.05$, and analyses were performed using SPSS 24 (IBM Corp). Cohen's d effect sizes (mean residual/SD) were calculated for performance parameters to classify the magnitude of estimation error as large (≥ 0.8), moderate (0.5–0.8), or small (< 0.5). All statistical analyses and graphical figures were conducted using R software (version 3.6.1; The R Foundation for Statistical Computing, Vienna, Austria), GraphPad (Prism 9), and JASP (v. 0.16.3).

Results

Performance

The performance data of the subjects recorded during the test period (simulated competition) are reported in Tables 1 and 2.

For the athletes (both men and women) in the CONT group, the average performance times for the 400-m races during D4, D5, and D6 were 55.71 ± 4.19 s (men: 53.32 ± 2.66 s; women: 59.90 ± 2.76 s), 55.78 ± 4.09 s (men: 53.58 ± 2.67 s; women: 59.62 ± 3.26 s), and 55.57 ± 4.34 s (men: 53.09 ± 2.68 s; women: 59.90 ± 3.02 s), respectively.

For the athletes (both men and women) in the RELAX group, the average performance times for the 400-m races during D4, D5, and D6 were 54.58 ± 6.85 s (men: 50.85 ± 2.50 s; women: 64.53 ± 2.92 s), 54.35 ± 6.65 s (men: 50.70 ± 2.09 s; women: 64.09 ± 3.20 s), and 54.42 ± 6.65 s (men: 50.74 ± 1.96 s; women: 64.23 ± 2.99 s), respectively. No significant differences were reported between the 2 conditions or between the different races for either condition ($p > 0.05$).

Lactatemia

The blood lactate values of the subjects during the test period (simulated competition) are reported in Tables 1 and 2. The mean maximum blood lactate concentrations for the subjects in the CONT group following D4, D5, and D6 were 20.0 ± 3.8 mmol, 19.8 ± 2.9 mmol, and 19.7 ± 1.4 mmol, respectively. In each case, peak lactate values were consistently reached at 3 minutes posttrace.

The mean maximum blood lactate concentrations for the subjects in the RELAX group following D4, D5, and D6 were 20.1 ± 2.1 mmol, 19.3 ± 1.3 mmol, and 19.3 ± 1.4 mmol, respectively. In each case, peak lactate values were consistently reached at 3 minutes posttrace. No significant differences were observed between the different maximum blood lactate concentrations or between the 2 conditions during the test period ($p > 0.05$).

Competitive State Anxiety Inventory-2

For the subjects in the CONT group, somatic anxiety significantly increased between familiarization and D4 ($p < 0.05$; familiarization: 9.42 ± 11.21 a.u.; D4: 10.30 ± 11.06 a.u.). Self-confidence in the CONT group significantly decreased between familiarization and D4 (familiarization: 13.45 ± 9.22 a.u.; D4: 12.24 ± 8.56 a.u.). For the subjects in the RELAX group, somatic anxiety significantly increased between familiarization and D4 ($p < 0.05$; baseline: 8.78 ± 11.49 a.u.; D4: 10.58 ± 10.50 a.u.).

Heart Rate Variability

All HRV results are presented in Tables 1 and 2 and illustrated in Figure 2 (Figure 2) as means and SDs. For subjects in the CONT group, the RMSSD in the supine position significantly decreased between D4 and D5 ($p = 0.008$ and $ES = 0.989$; RMSSD D4: 90.29 ± 21.26 ms⁻¹ vs. RMSSD D5: 79.58 ± 20.46 ms⁻¹), while no significant difference was observed in the simulated competition for the RELAX group ($p > 0.05$).

In the CONT group, resting HR measured in the morning in the supine position significantly increased between D4 and D5 ($p < 0.05$ and $ES = -0.664$; HR D4: 54.99 ± 7.45 bpm vs. HR D5: 59.80 ± 8.12 bpm). In the RELAX group, morning HR in the supine position significantly increased between D4 and D5 400-m races ($p < 0.001$ and $ES = -1.533$; HR D4: 52.71 ± 8.60 bpm vs. HR D5 400 m: 57.31 ± 7.39 bpm), as well as between the D4 and D6 ($p < 0.05$ and $ES = -0.783$; D4: 52.71 ± 8.60 bpm vs. HR D6: 56.83 ± 12.73 bpm). No significant differences were reported between the 2 conditions for RMSSD and HR ($p > 0.05$).

Salivary Samples

Salivary sampling data are reported in Tables 1 and 2.

Alpha-Amylase. In the CONT group, salivary alpha-amylase concentration significantly decreased between pretest period and D4 ($p < 0.05$ $ES = 0.694$; pretest CONT: 40.811 ± 22.422 U·ml⁻¹; men: 34.39 ± 11.58 U·ml⁻¹ and women: 52.05 ± 33.80 U·ml⁻¹ vs. D4: 35.021 ± 21.603 U·ml⁻¹ men: 29.43 ± 12.98 U·ml⁻¹ and women: 44.80 ± 31.91 U·ml⁻¹). No significant differences were observed in the RELAX group between the pretest period and the test days during the simulated competition ($p > 0.05$).

The morning salivary alpha-amylase concentration on D4 was significantly lower in the CONT group compared with the RELAX group ($p < 0.05$ & $ES = -0.896$; CONT D4: 35.021 ± 21.603 U·ml⁻¹ men: 34.39 ± 11.58 U·ml⁻¹ and women: 52.05 ± 33.80 U·ml⁻¹ vs. RELAX D4: 58.588 ± 18.010 U·ml⁻¹; men: 50.23 ± 21.19 U·ml⁻¹ and women: 62.16 ± 9.25 U·ml⁻¹).

Table 1
Summary table of data for the CONT group.*

	CONTROL				
	Pretest (D1 to D3)	Test: heats (D4)	Test: semifinal (D5)	Test: final (D6)	Posttest (D7)
Performance (s)					
Time	—	55.71 ± 4.19	55.78 ± 4.09	55.57 ± 4.34	—
Lactatemia					
Pre	1.4 ± 0.6	1.7 ± 0.5	1.7 ± 0.5	1.7 ± 0.4	—
Post 3 minutes	—	20.0 ± 3.8	19.8 ± 2.9	19.7 ± 1.4	—
Post 6 minutes	—	19.1 ± 2.7	19.0 ± 0.8	18.8 ± 2.2	—
CSAI-2R (a.u.)					
Somatic anxiety	9.42 ± 11.21	10.30 ± 11.06	10.30 ± 10.48	9.97 ± 11.20	—
Cognitive anxiety	8.79 ± 13.54	10.09 ± 11.74	9.58 ± 12.58	8.30 ± 11.88	—
Self-confidence	13.45 ± 9.22	12.24 ± 8.56	12.39 ± 9.60	12.15 ± 8.86	—
HRV (ms ⁻¹ ·bpm)					
RMSSD	84.21 ± 6.28	90.29 ± 21.26	79.58 ± 20.46	81.13 ± 52.60	89.37 ± 4.61
HR	56.51 ± 1.24	54.99 ± 7.45	59.80 ± 8.12	59.40 ± 6.00	55.75 ± 6.11
Alpha-amylase (U·ml ⁻¹)					
Morning	40.811 ± 22.422	35.021 ± 21.603	45.335 ± 18.481	51.825 ± 22.745	49.283 ± 37.473
Pre	—	146.799 ± 90.595	127.229 ± 74.406	118.515 ± 55.759	—
Post	—	217.937 ± 85.594	219.567 ± 122.401	240.265 ± 120.107	—
Cortisol (μg·dl ⁻¹)					
Morning	0.366 ± 0.155	0.310 ± 0.123	0.374 ± 0.181	0.249 ± 0.122	0.393 ± 0.253
Pre	—	0.308 ± 0.244	0.183 ± 0.151	0.290 ± 0.258	—
Post	—	0.796 ± 0.385	0.601 ± 0.277	0.494 ± 0.247	—
Testosterone (μg·dl ⁻¹)					
Morning	0.018 ± 0.007	0.017 ± 0.007	0.017 ± 0.006	0.016 ± 0.008	0.017 ± 0.007
Pre	—	0.013 ± 0.004	0.012 ± 0.004	0.013 ± 0.007	—
Post	—	0.022 ± 0.010	0.018 ± 0.008	0.019 ± 0.008	—
Ratio T/C					
Morning	0.065 ± 0.028	0.069 ± 0.044	0.068 ± 0.050	0.096 ± 0.141	0.064 ± 0.037
Pre	—	0.068 ± 0.063	0.096 ± 0.065	0.104 ± 0.085	—
Post	—	0.063 ± 0.066	0.043 ± 0.035	0.059 ± 0.058	—
IL1-beta (pg·ml ⁻¹)					
Morning	835.547 ± 646.202	928.133 ± 760.721	815.842 ± 569.373	796.131 ± 641.798	1,309.800 ± 837.504
Pre	—	174.848 ± 136.783	248.275 ± 222.740	209.961 ± 225.692	—
Post	—	430.706 ± 398.889	493.564 ± 455.784	615.314 ± 474.296	—

*Summary table of various heart rate variability (HRV) markers, salivary biomarkers, sleep parameters, CSAI-2R score, performance, and lactate concentration for the CONT group.

Cortisol. In the CONT group, the morning salivary cortisol concentration significantly decreased between D5 and D6 ($p < 0.05$ & $ES = 0.76$; day 5: $0.374 \pm 0.181 \mu\text{g}\cdot\text{dl}^{-1}$; men: $0.342 \pm 0.206 \mu\text{g}\cdot\text{dl}^{-1}$; women: $0.428 \pm 0.135 \mu\text{g}\cdot\text{dl}^{-1}$ vs. day 6: $0.249 \pm 0.122 \mu\text{g}\cdot\text{dl}^{-1}$; men: $0.236 \pm 0.143 \mu\text{g}\cdot\text{dl}^{-1}$; women: $0.271 \pm 0.089 \mu\text{g}\cdot\text{dl}^{-1}$). Postrace salivary cortisol concentrations significantly decreased between the D4 and D6 for the CONT group athletes ($p < 0.01$ & $d = 0.979$; day 4 postrace: $0.796 \pm 0.385 \mu\text{g}\cdot\text{dl}^{-1}$; men: $0.708 \pm 0.423/\text{dl}$; women: $0.948 \pm 0.298 \mu\text{g}\cdot\text{dl}^{-1}$ vs. day 6 postrace: $0.494 \pm 0.247 \mu\text{g}\cdot\text{dl}^{-1}$; men: $0.477 \pm 0.239/\text{dl}$; women: $0.523 \pm 0.297 \mu\text{g}\cdot\text{dl}^{-1}$) (Figure 3).

No significant differences were observed in the RELAX group across the test period ($p > 0.05$). No significant differences were reported between the 2 conditions in salivary cortisol concentration ($p > 0.05$).

Testosterone. In the CONT group, salivary testosterone concentration significantly decreased between the average pretest period and day 6 (the third competition day) ($p < 0.05$; pretest: $0.018 \pm 0.007 \mu\text{g}\cdot\text{dl}^{-1}$; men: $0.019 \pm 0.008 \mu\text{g}\cdot\text{dl}^{-1}$; women: $0.017 \pm 0.007 \mu\text{g}\cdot\text{dl}^{-1}$ vs. day 6: $0.016 \pm 0.008 \mu\text{g}\cdot\text{dl}^{-1}$; men: $0.013 \pm 0.007 \mu\text{g}\cdot\text{dl}^{-1}$; women: $0.018 \pm 0.010 \mu\text{g}\cdot\text{dl}^{-1}$).

In the RELAX group, salivary testosterone concentration significantly decreased between the average pretest and day 5 ($p < 0.05$ and $ES = 0.699$; pretest: $0.025 \pm 0.014 \mu\text{g}\cdot\text{dl}^{-1}$; men: $0.025 \pm 0.014 \mu\text{g}\cdot\text{dl}^{-1}$; women: $0.023 \pm 0.017 \mu\text{g}\cdot\text{dl}^{-1}$ vs. D5: $0.021 \pm$

$0.012 \mu\text{g}\cdot\text{dl}^{-1}$; men: $0.023 \pm 0.012 \mu\text{g}\cdot\text{dl}^{-1}$; women: $0.017 \pm 0.010 \mu\text{g}\cdot\text{dl}^{-1}$) day 6 ($p < 0.05$ and $ES = 0.695$; D6: $0.021 \pm 0.013 \mu\text{g}\cdot\text{dl}^{-1}$; men: $0.021 \pm 0.012 \mu\text{g}\cdot\text{dl}^{-1}$; women: $0.019 \pm 0.014 \mu\text{g}\cdot\text{dl}^{-1}$), and D7 ($p < 0.05$ and $ES = 0.742$; D7: $0.018 \pm 0.012 \mu\text{g}\cdot\text{dl}^{-1}$; men: $\pm \mu\text{g}\cdot\text{dl}^{-1}$; women: $0.021 \pm 0.012 \mu\text{g}\cdot\text{dl}^{-1}$).

Ratio T/C. For the T/C ratio, no significant differences were observed in the CONT and RELAX groups ($p > 0.05$).

IL1-β. In the CONT group, salivary IL1-β concentrations significantly increased between pretest and posttest period (D7) ($p < 0.05$ and $ES = -0.867$; pretest: $835.547 \pm 646.202 \text{ pg}\cdot\text{ml}^{-1}$; men: $802.819 \pm 659.051 \text{ pg}\cdot\text{ml}^{-1}$; women: $892.823 \pm 718.576 \text{ pg}\cdot\text{ml}^{-1}$ vs. D7: $1,309.800 \pm 837.504 \text{ pg}\cdot\text{ml}^{-1}$; men: $1,397.786 \pm 970.716 \text{ pg}\cdot\text{ml}^{-1}$; women: $1,578.375 \pm 1,028.434 \text{ pg}\cdot\text{ml}^{-1}$). IL-1β levels showed a significant postrace increase by day 7 (D7). No significant differences in IL1-β concentration were observed in the RELAX group ($p > 0.05$).

Sleep Markers

The sleep markers of the subjects recorded during the protocol are reported in Figure 1A and Table 2.

Total Sleep Duration. No significant differences were reported in the CONT group between the pretest period, the test period, and

Table 2
Summary table of data for the RELAX group.*

	RELAX				
	Pretest (D1 to D3)	Test: heats (D4)	Test: semifinal (D5)	Test: final (D6)	Posttest (D7)
Performance (s)					
Time	—	54.58 ± 6.85	54.35 ± 6.65	54.42 ± 6.65	—
Lactatemia (mmol·L ⁻¹)					
Pre	1.5 ± 0.3	1.6 ± 0.4	1.6 ± 0.4	1.7 ± 0.5	—
Post 3 minutes	—	20.1 ± 2.1	19.3 ± 1.3	19.3 ± 1.4	—
Post 6 minutes	—	18.4 ± 2.1	17.9 ± 3.0	17.6 ± 2.3	—
CSAI-2R (a.u.)					
Somatic anxiety	8.78 ± 11.49	10.58 ± 10.50	9.76 ± 11.32	9.55 ± 10.68	—
Cognitive anxiety	8.39 ± 12.69	9.52 ± 11.96	9.33 ± 11.88	8.97 ± 11.77	—
Self-confidence	14.15 ± 8.41	13.79 ± 8.45	13.18 ± 8.85	14.00 ± 8.61	—
HRV (ms ⁻¹ ·bpm)					
RMSSD	88.77 ± 37.29	97.61 ± 24.14	75.08 ± 35.50	79.33 ± 37.72	72.95 ± 35.22
HR	55.17 ± 9.66	52.71 ± 8.60	57.31 ± 7.39	56.83 ± 12.73	55.97 ± 8.32
Alpha-amylase (U·ml ⁻¹)					
Morning	47.759 ± 18.393	58.588 ± 18.010	46.523 ± 22.520	55.822 ± 29.517	51.317 ± 26.598
Pre	—	181.271 ± 97.518	212.089 ± 106.479	164.025 ± 90.457	—
Post	—	264.100 ± 87.989	280.336 ± 82.045	236.564 ± 61.877	—
Cortisol (μg·d ⁻¹)					
Morning	0.364 ± 0.147	0.375 ± 0.202	0.315 ± 0.140	0.332 ± 0.135	0.332 ± 0.137
Pre	—	0.249 ± 0.239	0.199 ± 0.096	0.235 ± 0.078	—
Post	—	0.641 ± 0.377	0.587 ± 0.371	0.502 ± 0.248	—
Testosterone (μg·dl ⁻¹)					
Morning	0.025 ± 0.014	0.025 ± 0.014	0.021 ± 0.012	0.021 ± 0.013	0.018 ± 0.012
Pre	—	0.017 ± 0.008	0.017 ± 0.008	0.018 ± 0.009	—
Post	—	0.024 ± 0.014	0.023 ± 0.011	0.026 ± 0.013	—
Ratio T/C					
Morning	0.084 ± 0.054	0.079 ± 0.044	0.074 ± 0.055	0.081 ± 0.057	0.052 ± 0.035
Pre	—	0.121 ± 0.109	0.134 ± 0.121	0.126 ± 0.132	—
Post	—	0.041 ± 0.040	0.045 ± 0.028	0.076 ± 0.085	—
IL1-beta (pg·ml ⁻¹)					
Morning	949.190 ± 597.011	1,315.445 ± 848.265	1,239.753 ± 901.914	785.930 ± 409.890	1,116.250 ± 837.102
Pre	—	321.798 ± 333.492	335.218 ± 207.184	293.569 ± 191.630	—
Post	—	506.478 ± 442.382	506.827 ± 525.667	507.536 ± 458.236	—

*Summary table of various heart rate variability (HRV) markers, salivary biomarkers.

the posttest period ($p > 0.05$). In the RELAX group, total sleep duration significantly increased between the pretest period and day 4 ($p < 0.05$; pretest: $24,555.2 \pm 4,697.0$ seconds; D4: $27,960.0 \pm 3,684.1$ seconds). Total sleep duration in the RELAX group decreased between D4 and the posttest period ($p < 0.05$; D4: $27,960.0 \pm 3,684.1$ seconds; D7: $24,174.0 \pm 5,221.18$ seconds) (Figure 4). Total sleep duration exhibited a significant increase in the RELAX group, with an average gain of approximately 56 minutes compared with baseline values ($p < 0.05$).

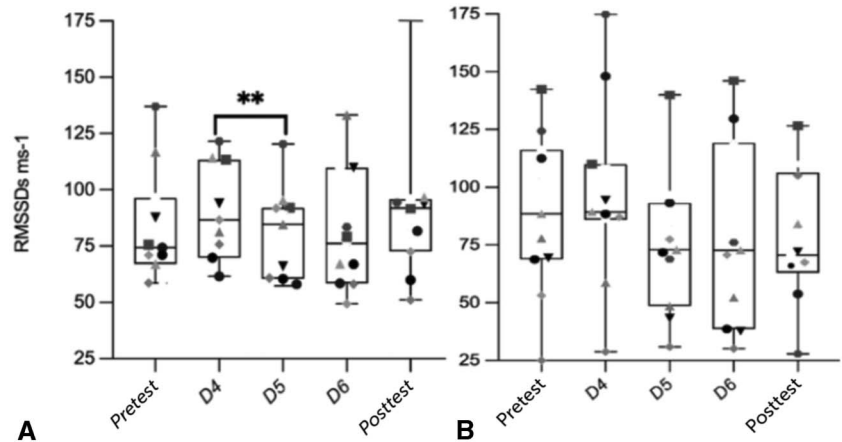


Figure 2. Evolution of the root mean square of successive differences (RMSSD) in the supine position obtained following an orthostatic test each morning of the protocol, with individual data for the CONT group (A) and the RELAX group (B). **Indicates a significant difference at $p < 0.01$.

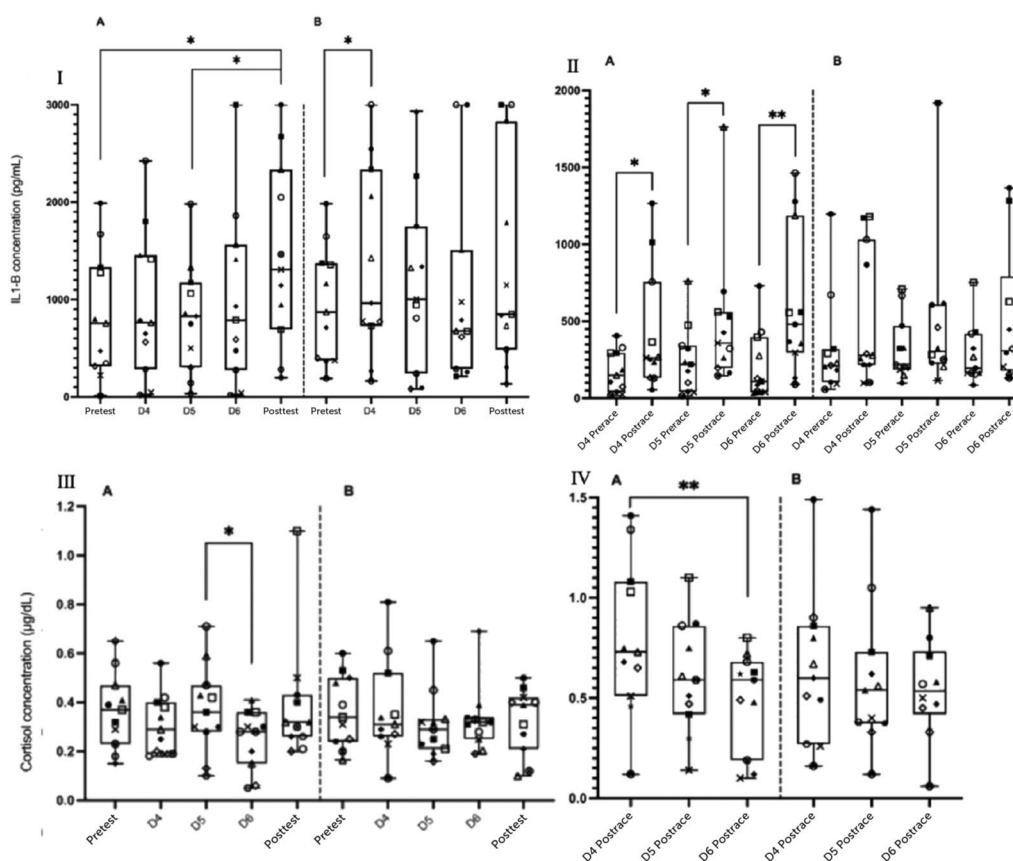


Figure 3. Measurements of IL-1- β and salivary cortisol with individual data for the CONT group (A) and the RELAX group (B). I and III: Evolution of IL-1- β and salivary cortisol each morning of the protocol. II and IV: Evolution of IL-1- β and salivary cortisol pretrace and posttrace each day of the testing period. *Indicates a significant difference at $p < 0.05$. **Indicates a significant difference at $p < 0.01$.

Rapid Eye Movement Sleep Duration. No significant difference was observed between the CONT and RELAX groups ($p > 0.05$).

Light Sleep Duration. No significant differences were observed for either the CONT or RELAX group ($p > 0.05$).

Deep Sleep Duration. No significant differences were observed in the CONT group ($p > 0.05$). In the RELAX group, deep sleep duration significantly increased between pretest and day 4 ($p < 0.05$; pretest: $8,772.18 \pm 2,391.3$ seconds; D4: $9,820.91 \pm 2,584.9$ seconds).

Sleep Latency. No significant differences were observed for any measurement between the CONT and RELAX groups ($p > 0.05$). However, when the percentage variance relative to the pretest period was analyzed, a significant difference was found on D5 between the CONT and RELAX groups ($p < 0.05$) (Figure 5).

Discussion

This study is the first to highlight the effects of a relaxation strategy using a systemic approach based on multiple biomarker analyses. The objective of this study was to evaluate the impact of coherent breathing on the regulation of psychophysiological stress in well-trained athletes during a simulated competition. The main results show that the practice of coherent breathing in the RELAX group improved parasympathetic activity, especially

RMSSD, known as a direct marker of vagal tone (9,26,39), stabilized salivary biomarkers such as alpha-amylase and cortisol, and increased the total sleep duration compared with the CONT group. Furthermore, the absence of significant modulation of IL-1- β in the RELAX group compared with the CONT group indicates a protective effect against inflammation responses induced by physical stress.

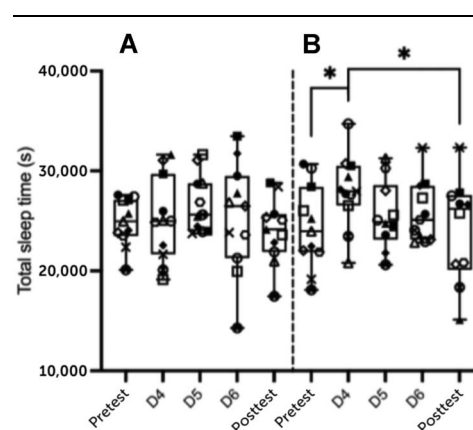


Figure 4. Evolution of total sleep time each night of the protocol, with individual data for the CONT group (A) and the RELAX group (B). *Indicates a significant difference at $p < 0.05$.

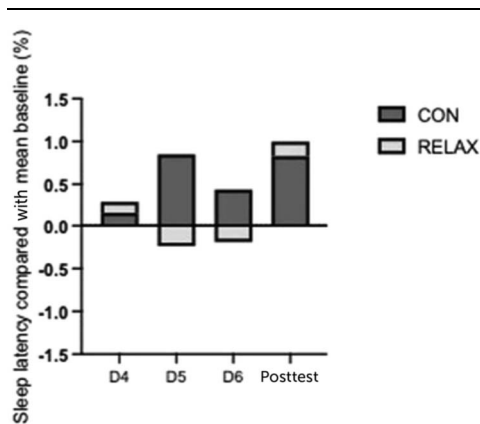


Figure 5. Evolution of sleep latency as a percentage (%) relative to baseline and each night of the protocol for the CONT group and the RELAX group.

In this study, performance times of specialized athletes across D4-D6 showed no significant differences between the CONT and RELAX groups ($p > 0.05$), indicating that coherent breathing did not acutely influence race outcomes. Similarly, posttrace blood lactate concentrations were consistent with values typically recorded in official competition, confirming that the simulated races successfully reproduced authentic physiological demands. The equivalence in both lactatemia and performance fatigue across conditions supports a valid comparison of the physiological and recovery responses, as both groups reached similar levels of exertion. Therefore, any differences observed in recovery metrics or autonomic regulation can be attributed specifically to the effects of cardiac coherence, rather than to disparities in physical effort or metabolic stress.

In contrast to the CONT group, which displayed a significant reduction in RMSSD, a key time-domain index of HRV (6,27), during simulated competition, athletes in the RELAX group preserved stable parasympathetic regulation (Figure 2). This divergence reflects a shift toward sympathetic dominance in the CONT group, a phenomenon frequently associated with competitive stress or repeated maximal effort (8,9). Coherent breathing mitigated this imbalance, supporting prior findings that highlight its modulatory effects on HRV, particularly on parasympathetic components (32,45). Although HRV is increasingly used to monitor recovery and training adaptation in athletes, its utility remains debated. Some authors emphasize its predictive capacity for overtraining and its relationship with vagal tone and stress resilience, whereas others caution against methodological variability and individual sensitivity to contextual factors (9,16). Nevertheless, consistent parasympathetic markers such as RMSSD remain valuable for assessing autonomic flexibility and recovery capacity, particularly in high-performance settings. Our results confirm that coherent breathing preserved these markers, suggesting a protective autonomic effect during competition-like conditions. To complement this analysis, we examined salivary alpha-amylase, an established biomarker of sympathetic nervous system activation and neuroendocrine stress. Its rapid responsiveness to sympathetic arousal makes it particularly relevant for evaluating competitive anxiety, as demonstrated in previous research (30). During the test period, the CONT group showed a significant increase in alpha-amylase levels, reflecting an acute psychophysiological stress response. By contrast, the RELAX group maintained more stable concentrations (Table 1), reinforcing the hypothesis that coherent breathing may dampen

excessive sympathetic activation—even if the precise neurophysiological pathways require further elucidation (23). The early divergence between groups (notably at D4) suggests a rapid modulatory effect of the breathing protocol on stress physiology. Mechanistically, coherent breathing restores autonomic balance through modulation of respiratory sinus arrhythmia: exhalation enhances parasympathetic activity (through vagal input), while inhalation promotes sympathetic drive (44). By synchronizing breathing with the intrinsic baroreflex frequency, coherent breathing amplifies parasympathetic tone through baroreflex resonance (36), enhancing cardiovascular flexibility. Taken together, these findings support the use of coherent breathing as a practical tool to regulate autonomic function and neuroendocrine stress in athletes exposed to repeated supramaximal effort. While its immediate effect on performance time may be limited, the stabilization of HRV and alpha-amylase levels indicates enhanced physiological resilience, an essential component of recovery and long-term training efficiency. This autonomic stabilization may also facilitate the effectiveness of recovery strategies such as sleep and the restoration of baseline homeostatic processes.

Furthermore, the results concerning salivary cortisol, a key biomarker of the stress response, support the potential benefits of coherent breathing in regulating the HPA axis. Two major observations emerged in the RELAX group practicing coherent breathing. First, a significant modulation of cortisol levels was observed between preexercise and postexercise periods, with consistent differences across the study timeline. This contrasts with the control group, which exhibited no notable modulation on the final day of competition. These findings suggest that coherent breathing may play a critical role in stabilizing the HPA axis, thereby preserving a balanced neuroendocrine response to repeated physical efforts. This phenomenon aligns with the concept of adrenal depletion (7), wherein prolonged exposure to intense stress results in diminished adrenal gland activity and reduced cortisol production. In the control group, the absence of a regulatory intervention such as coherent breathing appeared to lead to a progressive decline in cortisol production over the course of the exercises, indicating a reduced capacity to physiologically respond to repeated stressors. By contrast, athletes in the RELAX group maintained more consistent cortisol regulation, with consistent modulation throughout the competition days, suggesting that coherent breathing helped mitigate adrenal depletion.

However, when comparing precompetition cortisol levels between the CONT and RELAX (coherent breathing) groups, no significant differences were observed. This result aligns with previous studies, indicating that, in highly trained athletes, precompetitive cortisol levels may have a limited impact, particularly in contexts where stress exposure is controlled or moderate (51). Overall, coherent breathing appears to act as an effective modulator of both the ANS and HPA axis, enabling athletes to sustain stable cortisol levels despite the cumulative demands of competition. Moreover, the combined assessment of salivary alpha-amylase and cortisol may offer a robust approach for evaluating chronic stress, as it encompasses multiple pathways involved in psychophysiological stress responses. This dual-biomarker strategy allows for a more nuanced understanding of the temporal dynamics of stress-related physiological activation, as proposed in the scientific literature (1,20).

In line with this, and consistent with the previous findings, CSAI-2R scores reported for both conditions showed a significant increase in somatic and cognitive anxiety during the simulated competition. Moreover, the self-confidence of the CONT group

decreased significantly between familiarization and the first test day. This result is in line with previous studies indicating that a decline in self-confidence can negatively impact performance (18). Indeed, the relationships established by Craft et al. between cognitive anxiety, somatic anxiety, and self-confidence highlight that self-confidence is the only factor among the 3 that can positively influence performance. Thus, implementing such a breathing strategy could positively enhance self-confidence, potentially leading to improved performance over time (18). Considering the results presented earlier, our study demonstrates that coherent breathing attenuates psychophysiological stress responses by regulating ANS activity and the HPA axis. This breathing technique also appears to contribute to greater stress resilience and could be a valuable strategy for athletes participating in intense events, supporting effective stress regulation throughout the competition period.

Moreover, our study is the first to demonstrate the modulation of inflammatory biomarkers such as IL1- β following the implementation of heart coherence in athletes during a sequence of supramaximal exercises. Variations in salivary IL1- β concentration, a biomarker of the pro-inflammatory system, appear to be significantly influenced by coherent breathing. Two major observations emerged regarding IL1- β . First, a significant difference between baseline and posttest values was observed in the CONT group, whereas no such changes were found in the RELAX group. Second, significant variations between preexercise and postexercise measurements were noted throughout the experiment in the CONT group, which remained stable in the RELAX group. These findings, particularly the significant increase in IL1- β in the control group, are consistent with the scientific literature, where an enhanced inflammatory response is expected in reaction to repeated stress stimuli. However, the fact that the RELAX group showed no significant modulation of IL1- β , even after intense exercise, is particularly noteworthy. This is the first time that such an absence of an exercise-induced inflammatory response has been observed in connection with a breathing strategy (22). It suggests that coherent breathing may play a role in regulating the immune response, particularly by attenuating the activation of pro-inflammatory pathways like IL1- β . These findings highlight the potential anti-inflammatory and protective effect of coherent breathing against excessive increases in inflammatory cytokines in response to physical stress. The stabilization of IL1- β levels in the RELAX group shows that coherent breathing not only enhanced the regulation of the autonomic and neuroendocrine systems but also contributed to the modulation of immune activity. Thus, this practice could provide an effective strategy for preventing inflammatory overload and associated risks, particularly in contexts of high psychophysiological stress. In summary, coherent breathing has demonstrated positive effects on the regulation of neuroendocrine, autonomic, and immune functions during a sequence of supramaximal exercises in a simulated competition. Given the deleterious effects of psychophysiological stress on homeostasis (31) and the importance of sleep in this regulatory process, it is essential to consider the impact of such a breathing strategy on postexercise recovery and overall physiological balance.

As previously mentioned, the regulation of homeostasis is closely linked to both the quantity and quality of sleep. In response to stress, sleep plays a major role in homeostatic regulation processes, but it can also be negatively affected by homeostatic disturbances (49). Thus, sleep is a crucial variable in the recovery of athletes during competition. The results of our study confirm the positive impact of coherent breathing on sleep quantity and its association with the

regulation of the ANS and the HPA axis. Indeed, a significant increase of 56 minutes in total sleep duration was observed in the RELAX group on day 4 compared with baseline. This 15% improvement indicates enhanced sleep efficiency for athletes practicing coherent breathing. This phenomenon can be explained by improved autonomic regulation, as evidenced by the evolution of RMSSD and salivary alpha-amylase concentrations in the RELAX group during the testing period. Coherent breathing promotes parasympathetic activation, which is reflected in increased HRV, particularly through higher RMSSD values. Elevated HRV is strongly associated with better sleep quality and faster sleep onset, as it reflects a more adaptable and resilient ANS. As the links between parasympathetic stimulation and recovery have been established, sympathetic activity is linked with disruption of recovery processes, with the interplay of sympathetic activity products (cortisol & alpha-amylase) and sleep variables, as sleep latency and total deep sleep time. The practice of coherent breathing appears to have stabilized sympathetic nervous system activity, which likely facilitated an improvement in sleep quantity, mitigating the risk of sleep deprivation. These observations are consistent with Chauvineau's (2021) (11) findings, which highlight the importance of parasympathetic reactivation after physical exercise to improve sleep latency. A significant difference in sleep latency was also observed between the CONT and RELAX groups on day 5. The RELAX group showed reduced sleep latency, while the CONT group exhibited prolonged latency, supporting the hypothesis that coherent breathing practice helped maintain parasympathetic equilibrium. This parasympathetic modulation appears to play a crucial role in improving sleep onset and sleep quantity (5,49). Regarding the quality of sleep stages, significant differences were also observed in light and deep sleep duration. The RELAX group showed an increase in light sleep on day 4 compared with baseline, and an increase in deep sleep duration compared with the CONT group. Light sleep, corresponding to the first stage of non-REM sleep, is essential for cognitive restoration and memory consolidation (19). Deep sleep, characterized by high-amplitude delta waves, is crucial for physical recovery, neuroendocrine regulation, and long-term memory processing (19). By regulating ANS activity, coherent breathing also seems to have enhanced sleep architecture, promoting recovery during these essential recovery phases.

It is important to note that the results of our study were obtained from a population of athletes specialized in supramaximal efforts. Owing to the unique nature of the studied population, sleep monitoring was conducted using version 3 of the Oura Ring. However, data on sleep quality should be interpreted with caution, given known limitations in measurement accuracy between this tool and polysomnography. Indeed, an approximate margin of error of 10% was observed for all data related to sleep quality (stages), with only sleep quantity (including sleep latency and total sleep duration) having been validated against the gold standard (12,21). This protocol was conducted over 1 week and in the context of a simulated competition, rather than an official competitive event. Nonetheless, the experimental conditions were designed to closely mirror real-world athletic demands. Athletes engaged in a 3-race sequence over 3 consecutive days, comprising preliminary heats, semifinals, and a final, with direct confrontation between runners and standardized start procedures. This structure was designed to replicate the physiological and psychological stressors characteristic of high-stakes competitions. Notably, despite the simulated nature of the event, self-reported stress levels, blood lactate concentrations (28,33), and athletic performance metrics (33) were comparable to those typically observed in actual competitive settings. Furthermore, the timeframe for familiarization with coherent breathing may not have been optimal, as the chronic benefits of this strategy likely require longer

adaptation periods. Furthermore, as our protocol was conducted with athletes specialized in the 400 m during a simulated competition, it may be valuable to replicate this protocol in an actual competitive context.

In conclusion, our results show that coherent breathing has a beneficial effect on sleep regulation, particularly in sleep latency and total sleep duration. Although performance did not improve between the 2 conditions, this systemic approach, combining the analysis of ANS markers and neuroendocrine biomarkers with immune or confirms that practicing coherent breathing is an effective strategy for enhancing recovery in athletes subjected to repeated efforts and intense psychophysiological stress (50).

Practical Applications

Although research highlights the ability of elite athletes to overcome the effects of sleep deprivation during repeated supramaximal exercises, they are still exposed to a decline in psychological resources such as motivation, which plays a crucial role in maintaining performance levels (41). Indeed, as mentioned throughout our study, elite athletes face significant psychological stress due to the highly competitive environment (10), which can ultimately impact their ability to consistently achieve high-level performance. Some athletes are more sensitive than others to the psychophysiological stress induced by competition, and the disruptive effects of such stress on homeostasis can significantly affect recovery and final athletic performance. From this perspective, implementing strategies aimed at regulating the effects of stress appears to be a critical concern for high-level athletes engaged in international competitions. As described in our study, the practice of coherent breathing as a recovery strategy seems to be a highly promising intervention for this population. Given its simplicity, low associated cost, minimal time commitment (3 times 5 minutes per day), and demonstrated benefits for managing psychophysiological stress, performance professionals should encourage their athletes to adopt such a practice to enhance competitive performance. Thus, the use of coherent breathing in a competitive context could contribute to improved sleep quality, particularly for athletes most affected by stress. This breathing strategy would optimize recovery processes and support the sustained repetition of high-level performances within the tight constraints of competition schedules.

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